IEORE4004: Optimization Models & Methods **Fall 2025**

Introduction to Mathematical Optimization and Linear Programming

1 Formulating an optimization problem

A mathematical optimization problems is made of the following components. The first one is the **Data.** This is what is given to us in input, and cannot be changed. Depending on the problem, data could be the price of the goods we want to sell, the number of people working in our company, etc.

The other parts of an optimization problem are up to us to define – together, they give the **formulation**. We start with the **Decision variables**. They are not part of the data – actually, they are nowhere. It is up to us to define them. To determine which are the decision variables, ask yourself: which are the components of the problem that are not fixed, but are part of the solution?

Once we have settled on some decision variables, we should use them to formulate an **objective function**. What do we want to maximize or minimize? Usually, the objective function contains some profit and / or some cost associated to our problem. We should be able to express the objective function in terms of the decision variables discussed above. If not, we need to go back and change the decision variables.

Then we need to formulate the **constraints**. Constraints will tell us which solutions are actually feasible. For instance, if the decision variables describe the amount of products you are selling on the market, constraints may cap it as a function of your production means. Again, if we cannot describe all constraints in terms of the decision variables, we need to go back and change them.

Last, you may want to do a **sanity check**. To write our formulation, did we use all the information from the description of the problem? If not, either some of the information was redundant (unlikely, but it can happen), or our formulation did not incorporate all features of the problem (more likely). In the latter case, you have to go back to formulation, and think again whether it accurately describes your problem.

2 Linear Programming by examples

Example 1. (Production) A company produces two kinds of objects, chairs and tables:

- It takes 1 hour of work and 2 units of wood to build a table.
- It takes 1 hour of work and 1 unit of wood to build a chair.
- Each week, we have workforce for 80 hours and 100 units of wood.
- Each week, we can sell at most 40 tables, and as many chairs as we want.
- The profit for selling a table is \$3.
- The profit for selling a chair is \$2.

Oar goal is to decide the weekly production as to maximize the profit.

All of the above is the Data. Let's now formulate our problem step by step:

- Decisions variables. Since we need to decide how many chairs and tables to produce, we let:
  - x_1: number of tables we produce;
  - x_2: number of chairs we produce.
- Objective function. Given the profit information from the data, we want to maximize the following objective function:
$$3x_1 + 2x_2$$
- Constraints. First of all, we cannot produce a negative numbers of chairs or tables. Thus, we have
$$x_1, x_2 \geq 0.$$
These constraints are called nonnegativity and you will have to impose them for most problems. Then, we need to take into account the constraint on the amount of workforce available:
$$x_1 + x_2 \leq 80.$$
Similarly, we write a constraint that forces our production not to use more wood than the available amount:
$$2x_1 + x_2 \leq 100.$$
Last, we impose the constraints on the number of tables that we can sell:
$$x_1 \leq 40.$$

We can now do a sanity check and verify we have used all the information from the data (we did). This does not guarantee that our formulation is correct, however. We obtain therefore the formulation

$$\begin{array}{rrcrcl} \text{Max} & 3x_1 & + & 2x_2 & & \\ & x_1 & + & x_2 & \leq & 80 \\ & 2x_1 & + & x_2 & \leq & 100 \\ & x_1 & & & \leq & 40 \\ & & & x_1, x_2 & \geq & 0. \end{array}$$

The one above is an example of **Linear Program** (LP), since we want to maximize a linear objective function subject to a finite set of linear constraints. Here, *linear* means that all components of the problem are of the form

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n, \tag{1}$$

where $x_1, \ldots, x_n$ are the variables and $c_1, \ldots, c_n$ are numbers.

Note that to faithfully describe the problem, we would also need to impose that x_1, x_2 have integer values, since we cannot produce half a chair! These constraints would bring us outside the realm of linear programs (since $x_1 \in \mathbb{N}$ is not a linear constraint) and into that of **Integer Programs**. However, for the moment we will omit these constraints, and stay within the class of linear programs. The reason for this choice is that, as we will see later in the class, integer programs are much harder to solve.

2.1 Writing an LP using matrices and vectors

Let us now formulate the LP from Example 1 using matrices and vectors. In vector form, we often write (1) as

$$c^T x \quad \text{or} \quad cx \quad \text{or} \quad c \cdot x.$$

Similarly, we can write the constraints in matrix form as follows

$$\begin{array}{rcl} Ax & \leq & b \\ x & \geq & 0 \end{array}, \qquad \text{where } A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 80 \\ 100 \\ 40 \end{pmatrix}.$$

Thus, the LP from Example 1 can be written as

$$\max c^T x \ : \ Ax \leq b, \, x \geq 0. \tag{2}$$

Note that the formulation (2) is very general. Indeed, even an LP in minimization form can be written in maximization form, since we can rewrite

$$\min c^T x \ : \ Ax \leq b, \, x \geq 0.$$

in the following equivalent[1] form

$$-\max(-c)^T x \ : \ Ax \leq b, \, x \geq 0.$$

Moreover, constraints of the form

$$a^T x \geq \beta$$

can be written as

$$(-a^T)x \leq -\beta,$$

while constraints of the form

$$a^T x = \beta$$

can be written as the pair of constraints

$$a^T x \leq \beta, \quad (-a^T)x \leq -\beta.$$

(You are encouraged to try these reformulations on explicit examples.)

2.2 More examples and non-examples

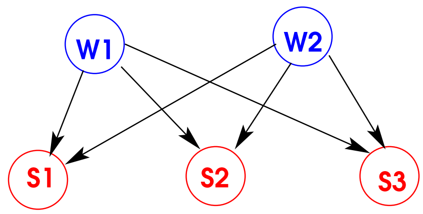
Example 2. (Transportation) A paper company wants to ship truckloads of paper from warehouses to stores.

- There are two warehouses with supplies of 100 and 200 units resp.
- There are three stores with demands 50, 100, 150 units resp.
- Per-unit transportation costs:

	S1	S2	S3
W1	10	12	14
W2	11	12	13

- Problem: minimize total transportation cost.

[1] Here equivalent means that the set of the optimal solution will be the same.



To formulate it as an LP, we let x_{ij} be the number of units sent between warehouse i ($i = 1,2$) and store j ($j = 1,2,3$). With these variables, you should be able to write the objective function and constraints, and obtain

$$\begin{aligned}
&\min\ 10x_{11} + 12x_{12} + 14x_{13} + 11x_{21} + 12x_{22} + 13x_{23} \\
&\text{Subject to} \\
&\quad x_{11} + x_{12} + x_{13} \ \leq\ 100 \\
&\quad x_{21} + x_{22} + x_{23} \ \leq\ 200 \\
&\quad x_{11} + x_{21} \ \geq\ 50 \\
&\quad x_{12} + x_{22} \ \geq\ 100 \\
&\quad x_{13} + x_{23} \ \geq\ 150 \\
&\quad x \geq 0
\end{aligned}$$

Note that we again omit to impose that variables are integral. Thus, we have again obtained a **Linear Program**. As an exercise, you may want to reduce your problem to the form (2).

Example 3. (Linear regression)

- We are given vectors $x_1, x_2, \ldots, x_N \in \mathbb{R}^n$.
- We are given scalar values $y_1, y_2, \ldots, y_N$.
- We want to come up with an "approximate" **linear** model for the data. We want to find vectors $a \in \mathbb{R}^n$, $b \in \mathbb{R}$ so that
$$y_i \ \approx\ a^T x_i + b, \quad 1 \leq i \leq N.$$

(So beware! Our variables are a, b, while data is given by $x_1, \ldots, x_N$ and $y_1, \ldots, y_N$). We therefore define the **error** to be
$$|a\bar{x} + b - \bar{y}|.$$

So our goal is to find a, b leading to small errors.

Method 1. (Most common) Linear regression.
$$\min_{a,b} \left\{ \sum_{i=1}^{N} (a^T x_i + b - y_i)^2 \right\}.$$

With the objective function above, we aim at minimizing the sums of the squares of the errors. This is clearly **not a linear program** (since we are squaring variables). However, it is formulated as to be a special (and tractable) kind of non-linear program, called **convex program**. We will see more about these problems later in the class.

Method 2. (Non-traditional method)
$$\min_{a,b} \max_{1 \leq i \leq N} |a^T x_i + b - y_i|.$$

With he objective function above, we aim at minimizing the largest error. Said otherwise, we choose a, b so that all errors are reasonably small. Although this does not look like a linear program (since it contains an absolute value)...it can be reformulated as a linear program! Let's do that in two steps.

$$\begin{aligned}
&\min\ V && (3)\\
&\text{Subject to} && (4)\\
&V \ \geq\ |a^T x_i + b - y_i|, \qquad 1 \leq i \leq N. && (5)
\end{aligned}$$

Here the **variables** are V, a (both n-vectors) and b. But is this really linear? No: (5) is not linear.

Second attempt:

$$\begin{aligned}
&\min\ V && (6)\\
&\text{Subject to} && (7)\\
&V \ \geq\ a^T x_i + b - y_i, \qquad 1 \leq i \leq N, && (8)\\
&V \ \geq\ -a^T x_i - b + y_i, \qquad 1 \leq i \leq N. && (9)
\end{aligned}$$

This is linear!

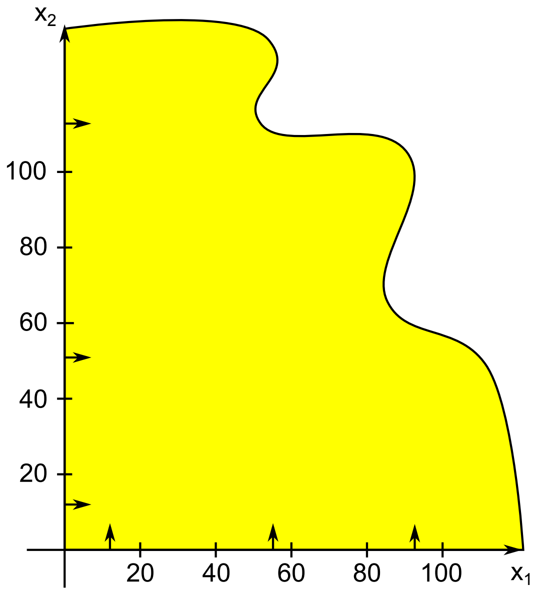
The graphical method

We will now show a technique to solve any 2-dimensional linear program (i.e., with 2 variables).

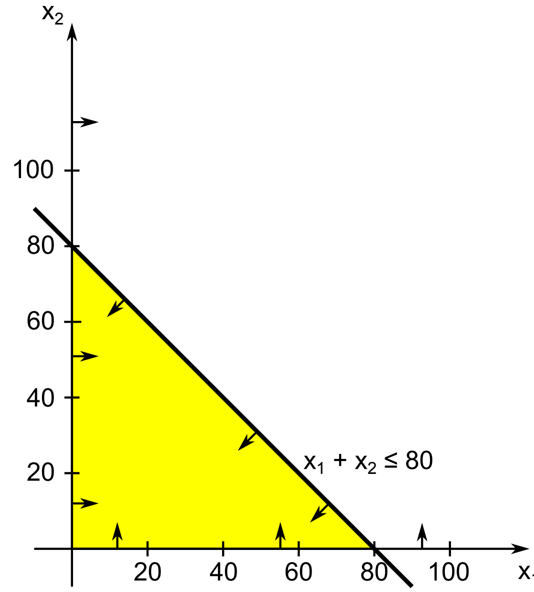
$$\begin{array}{lrcrcl}
\text{Max} & 3x_1 & + & 2x_2 & & \\
 & x_1 & + & x_2 & \leq & 80 \\
 & 2x_1 & + & x_2 & \leq & 100 \\
 & x_1 & & & \leq & 40 \\
 & & & x_1, x_2 & \geq & 0
\end{array}$$

1. Find the feasible region.

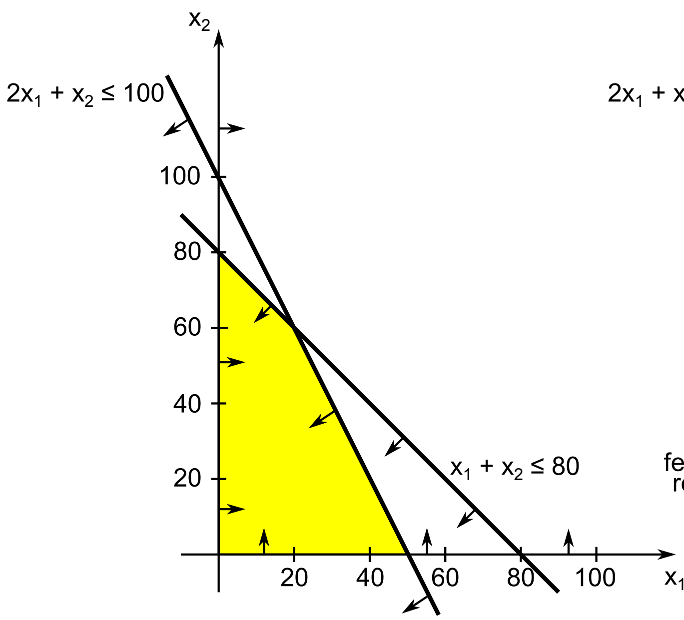
   - Plot each constraint as an equation $\equiv$ line in the plane
   - Feasible points on one side of the line – plug in (0,0) to find out which



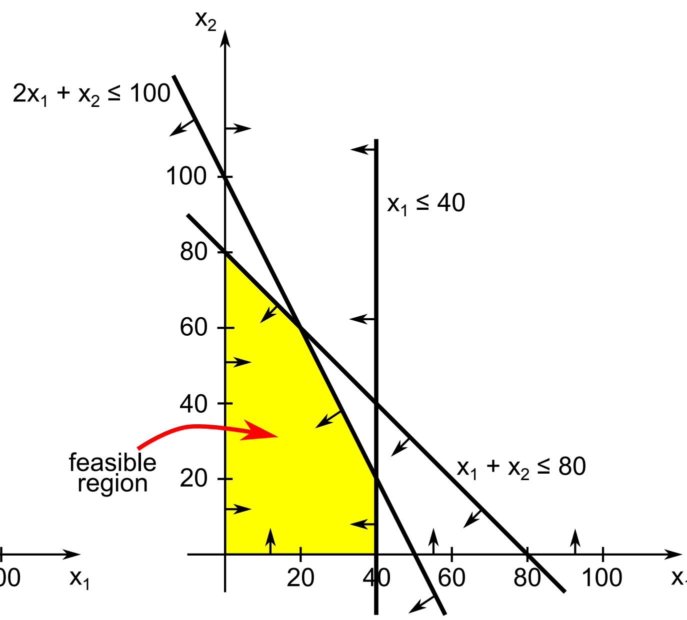
Start with $x_1 \geq 0$ and $x_2 \geq 0$



add $x_1 + x_2 \leq 80$



add $2x_1 + x_2 \leq 100$

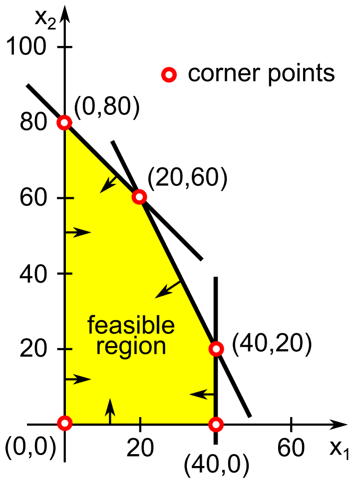


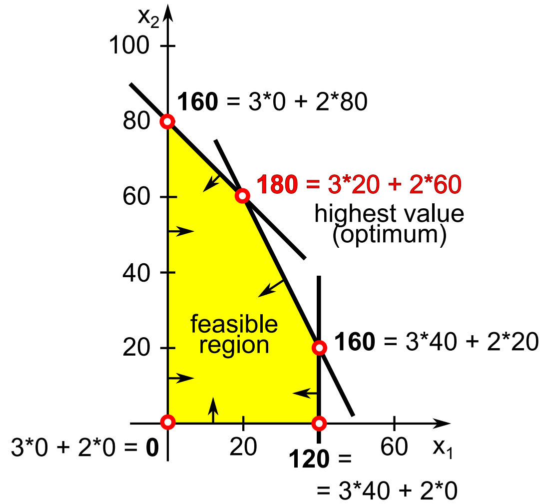
add $x_1 \leq 40$

A **corner** (extreme) point X of the region $R \equiv$ every line through X intersects R in a segment whose one endpoint is X. Solving a linear program amounts to finding a best corner point by the following theorem.

Theorem 1. *If a linear program with nonnegative variables has an* **optimal** *solution, then it also has an* **optimal solution** *that is a* **corner point** *of the feasible region.*

Exercise. Try to find all corner points. Evaluate the objective function $3x_1 + 2x_2$ at those points.





Problem: there may be too many corner points to check. There's a better way.

Iso-value line $\equiv$ in all points on this line the objective function has the same value

For our objective $3x_1 + 2x_2$ an iso-value line consists of points satisfying $3x_1 + 2x_2 = z$ where z is some number.

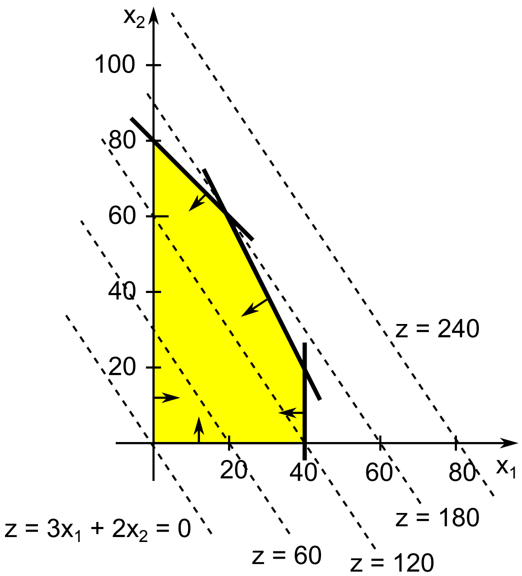
Main steps of the **Graphical Method**:

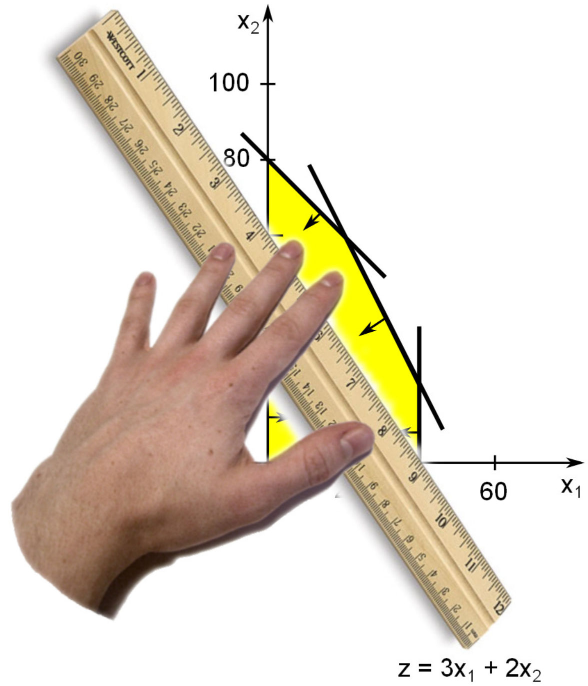
1. Find the feasible region.

2. Plot an iso-value (isoprofit, isocost) line for some value.

3. Slide the line in the direction of increasing value until it only touches the region.

4. Read-off an optimal solution.





Optimal solution is $(x_1, x_2) = (20, 60)$.

Observe that this point is the intersection of two lines forming the boundary of the feasible region. Recall that lines we use to construct the feasible region come from inequalities (the points on the line satisfy the particular inequality with equality). The next figure highlights the situation at the optimum.